

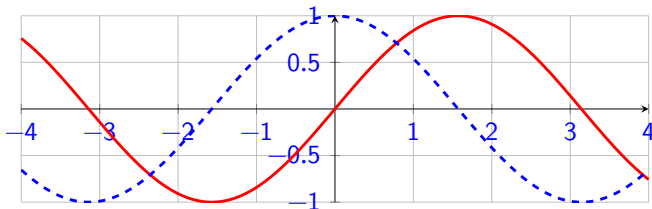
2627-MA140: Week 03, Lecture 1 (L07)

2627-MA140: Week 03, Lecture 1 (L07) The Squeeze Theorem

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University of Galway

Tuesday, 22 September, 2026



Reminder

- ▶ **Problem Set 1** has a deadline of 5pm, Monday 28 September. You can access it on Canvas... 2628-MA140... Assignments... [Problem Set 1](#). Click on [Problem 1](#) (e.g.) to get started.
- ▶ The **Tutorial Sheet** is available at <https://universityofgalway.instructure.com/courses/58477/files/3810601>
- ▶ Assignment 2 will open by Friday.

The first of the two class tests will take place 2 weeks from now:

Tuesday, 6th October.

- ▶ You will have 40 minutes to complete the test, which will be composed of multiple choice questions.
- ▶ Main venues: ENG-G017 or ENG-G018 (you'll be told which)
- ▶ I need to gather information on Reasonable Accommodations for tests. If you want to avail of such, please complete this form: <https://forms.cloud.microsoft/e/rBiq8RHHY2> by **10am, Tuesday 29 September.**



Topics we'll squeeze into today's class:

- 1 The Squeeze Theorem
- 2 $\sin(\theta)/\theta$
 - Some trig facts
 - “Log tables”
 - The proof
- 3 Other examples
- 4 Exercises

For more, see Chapter 2 (Limits) of Strang and Herman's **Calculus**, especially Sections 2.3 (Limit Laws) and 2.2 (Limit of a Function).

The Squeeze Theorem

Last week we learned about the formal definition of a limit, and the “Limit Laws” for calculating the limit of certain functions, such as rational functions.

We’ll now study another method for computing the limit of a (complicated) function, by “trapping” it between two simpler functions.

The Squeeze Theorem

The Squeeze Theorem (a.k.a. Sandwich Theorem)

Suppose that we have three functions f , g and h on some interval $[x_0, x_1]$, with

$$g(x) \leq f(x) \leq h(x),$$

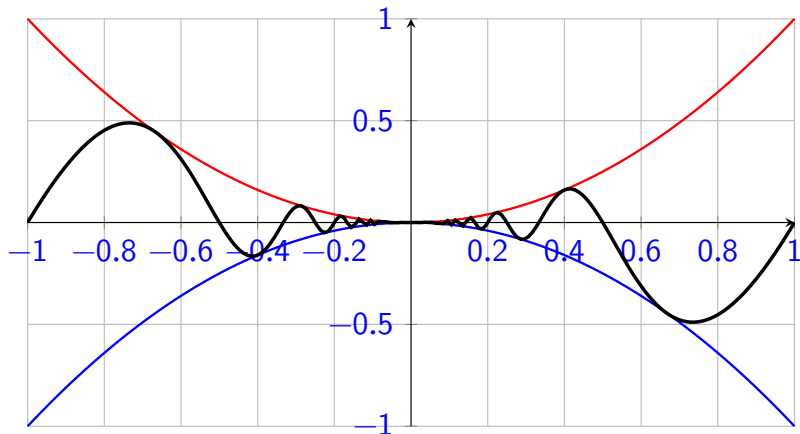
and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L,$$

for some $a \in [x_0, x_1]$. Then $\lim_{x \rightarrow a} f(x) = L$.

That is: if $g(x)$ and $h(x)$ have the same limit as $x \rightarrow a$, and $f(x)$ is “squeezed” between them, then so too does $f(x)$.

The Squeeze Theorem



The Squeeze Theorem

Example

Suppose $f(x)$ is a function such that

$$1 - \frac{x^2}{4} \leq f(x) \leq 1 + \frac{x^2}{2}, \text{ for all } x \neq 0$$

Find $\lim_{x \rightarrow 0} f(x)$.

$$\sin(\theta)/\theta$$

Next, we will use the Squeeze Theorem to explain **an important limit**:

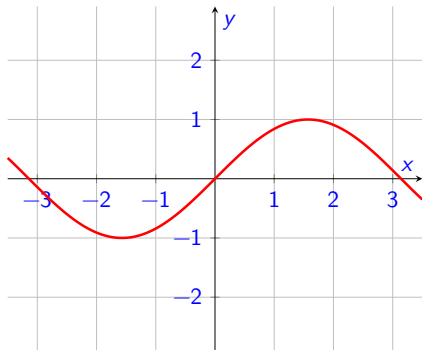
$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

First: check that we can't just evaluate $f(\theta) = \frac{\sin(\theta)}{\theta}$ at $\theta = 0$.

$$\sin(\theta)/\theta$$

Next, First, let's just convince ourselves that it is likely that

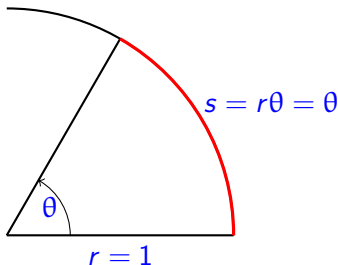
$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$



Next, we review some facts about trigonometric functions.

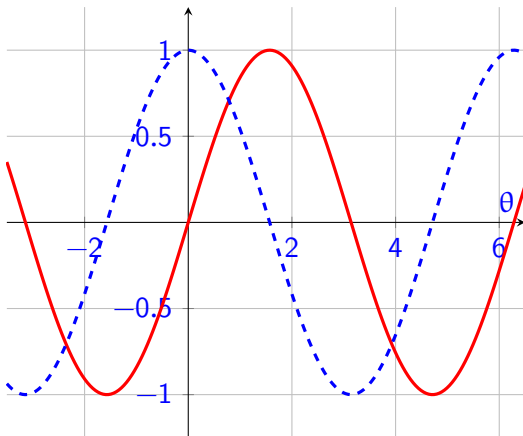
- ▶ **In this module, we only every use radians** (never degrees).
- ▶ Given a right-angled triangle, with sides a , b and h (the hypotenuse), $\sin \theta = \frac{b}{h}$, $\cos \theta = \frac{a}{h}$, $\tan \theta = \frac{b}{a} = \frac{\sin \theta}{\cos \theta}$

- ▶ The length of a sector, subtended by the angle θ , of a circle of radius r , is $s = r\theta$. In particular, for the unit circle **the angle (in radians) is the length of the arc..**



- ▶ Area of a sector of a circle is $\frac{1}{2}r^2\theta$ where r is the radius of the circle, and θ is the angle subtended by the sector.

- The $\sin$ (red) and $\cos$ (blue) functions look like this:

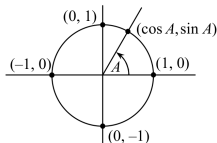


Various other facts are summarised in the State Examination Commission’s Tables:

Triantánacht

$$\tan A = \frac{\sin A}{\cos A} \quad \cot A = \frac{\cos A}{\sin A}$$

$$\sec A = \frac{1}{\cos A} \quad \operatorname{cosec} A = \frac{1}{\sin A}$$



Trigonometry

$$\cos^2 A + \sin^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\cos(-A) = \cos A$$

$$\sin(-A) = -\sin A$$

$$\tan(-A) = -\tan A$$

Nóta: Bíonn $\tan A$ agus $\sec A$ gan sainiú nuair $\cos A = 0$.
Bíonn $\cot A$ agus $\operatorname{cosec} A$ gan sainiú nuair $\sin A = 0$.

Note: $\tan A$ and $\sec A$ are not defined when $\cos A = 0$.
 $\cot A$ and $\operatorname{cosec} A$ are not defined when $\sin A = 0$.

A (céimeanna)	0°	90°	180°	270°	30°	45°	60°	A (degrees)
A (raidian)	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	A (radians)
$\cos A$	1	0	-1	0	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\cos A$
$\sin A$	0	1	0	-1	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	$\sin A$
$\tan A$	0	-	0	-	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\tan A$

$$1 \text{ rad.} \approx 57.296^\circ$$

$$1^\circ \approx 0.01745 \text{ rad.}$$

Foirmlí uillinneacha comhshuite

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Compound angle formulae

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Foirmlí uillinneacha dúbailte

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos^2 A = \frac{1}{2}(1 + \cos 2A)$$

$$\sin^2 A = \frac{1}{2}(1 - \cos 2A)$$

Double angle formulae

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$$

Iolraigh a thiontú ina suimeanna agus ina ndifríochtaí**Products to sums and differences**

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

Suimeanna agus difríochtaí a thiontú ina n-iolraigh**Sums and differences to products**

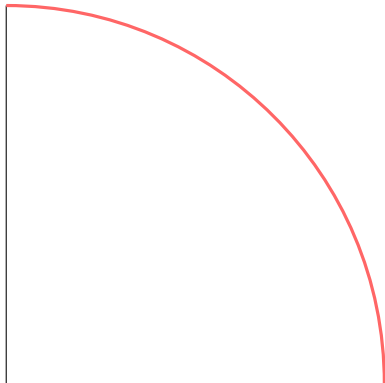
$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

To show that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, consider the quadrant of the unit circle shown below.



Other examples

Example

Evaluate $\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(2x)}$

Other examples

Example

Evaluate $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2}$

Exercise 3.1.1

Suppose that $g(x) = 9x^2 - 3x + 1/4$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x .

1. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1/3} f(x)$? If so, do so. If not, explain why.
2. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1/6} f(x)$? If so, do so. If not, explain why.

Exercises

Exercise 3.1.2 (from 2425-MA140 Exam)

Suppose that $g(x) = 2x^4 + x^2$, and $f(x)$ is such that $-g(x) \leq f(x) \leq g(x)$ for all x .

1. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 0} f(x)$? If so, do so. If not, explain why.
2. Can one use the Squeeze Theorem to determine $\lim_{x \rightarrow 1} f(x)$? If so, do so. If not, explain why.

Exercise 3.1.3 (from 2023/24 Q1(b))

Evaluate

$$\lim_{\theta \rightarrow 0} \frac{2 \sin(\theta)}{\theta + 3 \tan(\theta)}$$